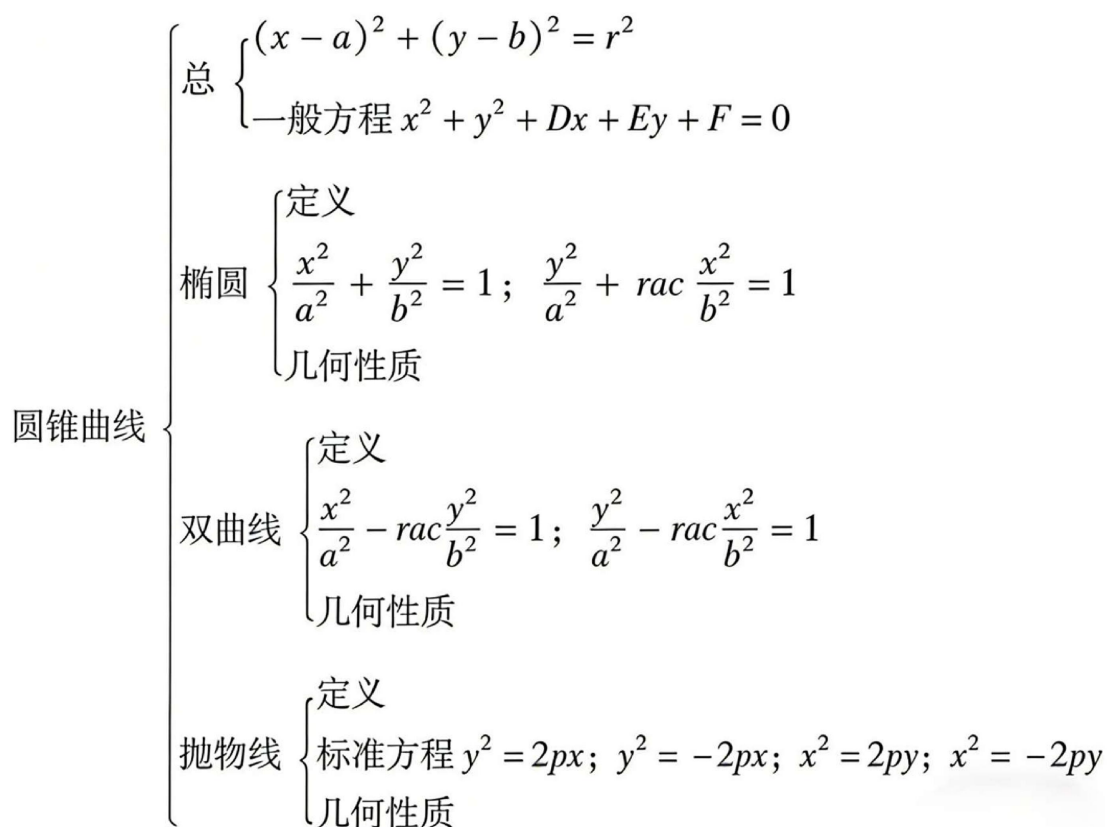


圆锥曲线知识点详解与应用

(中英文对照版)

第八章 圆锥曲线知识点结构图



8.1 圆(Circle)

一、基础知识(Basic Knowledge)

1. 定义(Definition)

平面内到定点(圆心)的距离等于定长(半径)的所有点的集合,称为圆。

(The set of all points in a plane that are equidistant from a fixed point (center) to a fixed length (radius) is called a circle.)

2. 核心公式(Core Formulas)

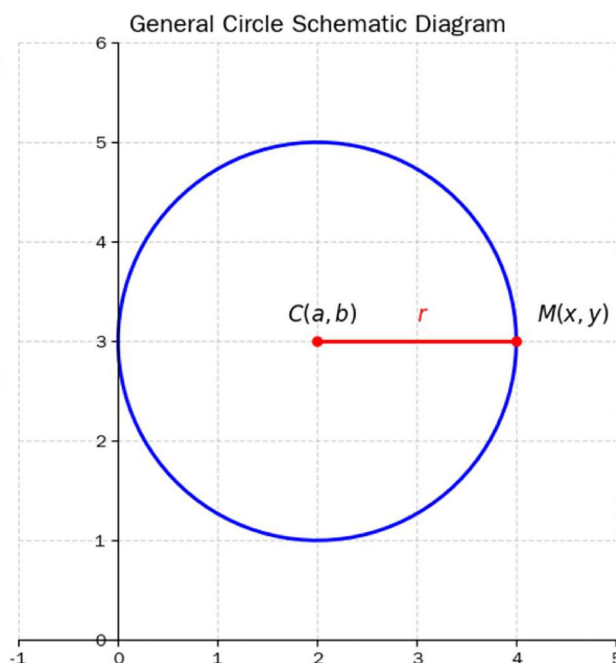
- 标准方程(Standard Equation): 设圆心为 $C(a, b)$, 半径为 r , 则圆的标准方程为

(Let the center be $C(a, b)$ and the radius be r , then the standard equation of the circle is): $(x - a)^2 + (y - b)^2 = r^2 (r > 0)$

- 一般方程(General Equation): $x^2 + y^2 + Dx + Ey + F = 0 (D^2 + E^2 - 4F > 0)$ 对应圆心(Corresponding center): $(-\frac{D}{2}, -\frac{E}{2})$, 半径(Radius): $r = \frac{1}{2}\sqrt{D^2 + E^2 - 4F}$

3. 公式推导(Formula Derivation)

- 标准方程推导(Derivation of Standard Equation): 设圆上任意一点 $M(x, y)$, 圆心 $C(a, b)$, 根据定义(According to the definition), $|MC| = r$ 。由两点间距离公式(From the distance formula between two points): $\sqrt{(x - a)^2 + (y - b)^2}$, 两边平方得(Square both sides): $(x - a)^2 + (y - b)^2 = r^2$, 即标准方程(which is the standard equation)。



- 一般方程推导(Derivation of General Equation): 将标准方程展开(Expand the standard equation): $x^2 - 2ax + a^2 + y^2 - 2by + b^2 - r^2 = 0$, 整理为(arrange as):

$x^2 + y^2 - 2ax - 2by + (a^2 + b^2 - r^2) = 0$ 。令 $D = -2a, E = -2b, F = a^2 + b^2 - r^2$, 则得一般方程(then the general equation is obtained) $x^2 + y^2 + Dx + Ey + F = 0$ 。反向推导圆心和半径(Reverse derivation of center and radius): 由 $D = -2a$ 得 $a = -\frac{D}{2}, E = -2b$ 得 $b = -\frac{E}{2}$; 代入 $F = a^2 + b^2 - r^2$, 解得(solve for) $r = \sqrt{a^2 + b^2 - F} = \frac{1}{2}\sqrt{D^2 + E^2 - 4F}$, 且需满足(and must satisfy) $r > 0$, 故 $D^2 + E^2 - 4F > 0$ (否则方程无实数解, 不表示圆)(otherwise the equation has no real solutions and does not represent a circle)。

二、例题(Examples)

例题 1(Example 1)

求以 $C(1, 3)$ 为圆心, 半径是 $\sqrt{5}$ 的圆的方程。(Find the equation of the circle with center $C(1, 3)$ and radius $\sqrt{5}$.)

解析(Analysis): 直接利用标准方程(Directly use the standard equation), 圆心 $a = 1, b = 3$, 半径 $r = \sqrt{5}$, 则 $r^2 = 5$ 。

代入标准方程得(Substitute into the standard equation): $(x - 1)^2 + (y - 3)^2 = 5$ 。

例题 2(Example 2)

圆的方程是 $(x - 3)^2 + (y + 2)^2 = 4$, 求圆心和半径。(The equation of a circle is $(x - 3)^2 + (y + 2)^2 = 4$, find the center and radius.)

解析(Analysis): 将方程与标准形式(Compare the equation with the standard form) $(x - a)^2 + (y - b)^2 = r^2$ 对比(compare), 可得(we get):

$a = 3, b = -2$ (注意(note) $y + 2 = y - (-2)$), $r^2 = 4$, 故 $r = 2$ 。

因此, 圆心为(Therefore, the center is) $(3, -2)$, 半径为(and the radius is) 2。

例题 3(Example 3)

圆的一般方程是 $x^2 + y^2 + 6x - 4y - 3 = 0$, 求圆心和半径。(The general equation of a circle is $x^2 + y^2 + 6x - 4y - 3 = 0$, find the center and radius.)

解析(Analysis): 方法一(公式法)(Method 1 (Formula Method)): 对比一般方程(Compare with the general equation) $x^2 + y^2 + Dx + Ey + F = 0$, 得(we get) $D = 6, E = -4, F = -3$ 。

圆心(Center): $\left(-\frac{6}{2}, -\frac{-4}{2}\right) = (-3, 2)$;

半径(Radius): $r = \frac{1}{2}\sqrt{6^2 + (-4)^2 - 4 \times (-3)} = \frac{1}{2}\sqrt{36 + 16 + 12} = \frac{1}{2}\sqrt{64} = 4$ 。

方法二(配方法)(Method 2 (Completing the Square Method)): 将一般方程配方(Complete the square for the general equation):

$$x^2 + 6x + 9 + y^2 - 4y + 4 = 3 + 9 + 4, \text{即(that is)} (x + 3)^2 + (y - 2)^2 = 16,$$

对比标准方程(Compare with the standard equation), 圆心(center) $(-3, 2)$, 半径(radius) 4 (两种方法相互验证, 提高准确性)(The two methods verify each other to improve accuracy)。

三、易错点解析(Common Mistakes Analysis)

1. 忽略一般方程的成立条件(Ignoring the establishment condition of the general equation): $D^2 + E^2 - 4F > 0$, 若该条件不满足, 方程不表示圆(可能表示点或无轨迹), 解题时需先验证(If this condition is not satisfied, the equation does not represent a circle (it may represent a point or no trajectory), and it needs to be verified first when solving problems)。
2. 标准方程中圆心坐标符号错误(Sign error of the center coordinates in the standard equation): 如方程 $(x + 1)^2 + (y - 2)^2 = r^2$, 圆心是 $(-1, 2)$, 而非 $(1, 2)$, 需注意“ $x - a$ ”中 a 是圆心横坐标的相反数(For example, in the equation $(x + 1)^2 + (y - 2)^2 = r^2$, the center is $(-1, 2)$ instead of $(1, 2)$. Note that a in “ $x - a$ ” is the opposite of the abscissa of the center)。
3. 半径计算遗漏根号或系数(Missing square root or coefficient in radius calculation): 一般方程的半径是 $\frac{1}{2}\sqrt{D^2 + E^2 - 4F}$, 容易忘记乘以 $\frac{1}{2}$ 或遗漏根号(The radius of the general equation is $\frac{1}{2}\sqrt{D^2 + E^2 - 4F}$, and it is easy to forget to multiply by $\frac{1}{2}$ or miss the square root)。

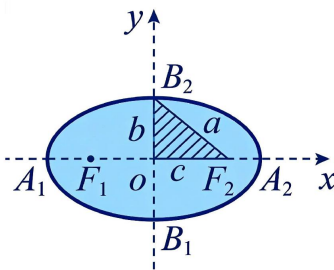
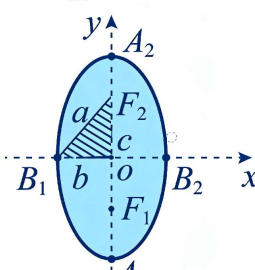
8.2 椭圆(Ellipse)

一、基础知识(Basic Knowledge)

1. 定义(Definition)

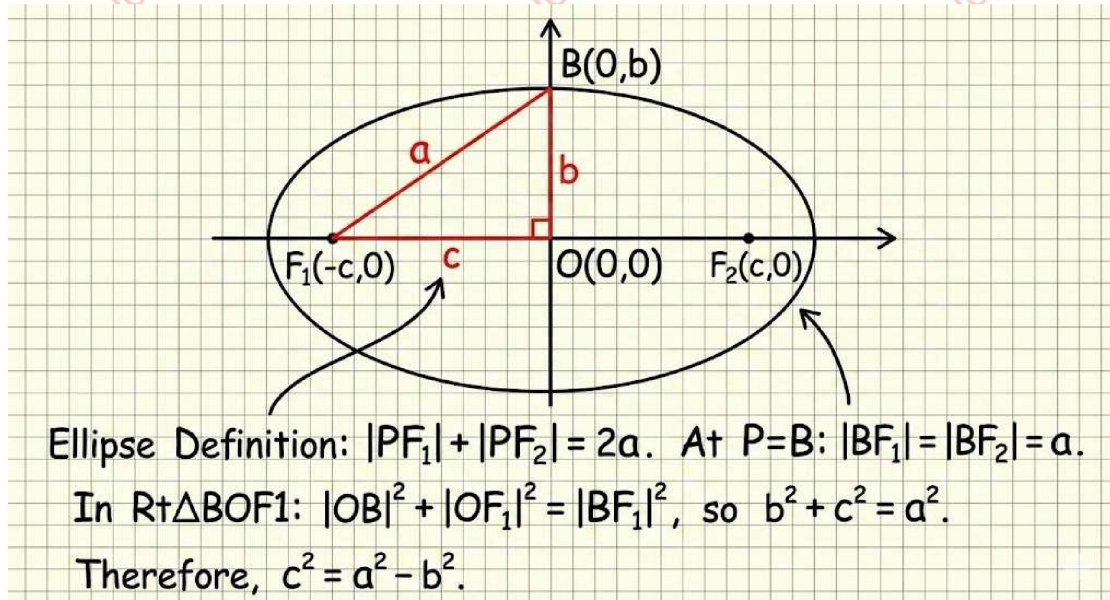
平面内与两个定点 F_1 、 F_2 (焦点)的距离之和等于常数 $2a(2a > |F_1F_2| = 2c, a > c > 0)$ 的点的轨迹,称为椭圆。(The set of all points in a plane where the sum of the distances from two fixed points F_1 and F_2 (foci) is equal to a constant $2a$ ($2a > |F_1F_2| = 2c, a > c > 0$) is called an ellipse.)

2. 核心公式与几何性质(Core Formulas and Geometric Properties)

标准方程 (Standard Equation)	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b > 0)$	$\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1 (a > b > 0)$
		
a, b, c 关系 (Relationship between a, b, c)	$c^2 = a^2 - b^2$ (a : 长半轴(semi-major axis), b : 短半轴(semi-minor axis), c : 半焦距(semi-focal length))	
范围(Range)	$-a \leq x \leq a, -b \leq y \leq b$	$-b \leq x \leq b, -a \leq y \leq a$
中心(Center)	$(0, 0)$ (原点(origin))	
对称轴 (Axis of Symmetry)	x 轴和 y 轴是对称轴,长轴长 $2a$,短轴长 $2b$	
顶点(Vertices)	$(\pm a, 0), (0, \pm b)$	$(0, \pm a), (\pm b, 0)$
焦点(Foci)	$F_1(-c, 0), F_2(c, 0)$	$F_1(0, -c), F_2(0, c)$
焦距(Focal Length)	$ F_1F_2 = 2c (c > 0)$	
离心率(Eccentricity)	$e = \frac{c}{a} (0 < e < 1)$ e 越小椭圆越圆(the smaller e is, the more circular the ellipse is), e 越大椭圆越扁(the larger e is, the flatter the ellipse is)	

3. 公式推导(Formula Derivation)

- $c^2 = a^2 - b^2$ 推导 (Derivation of $c^2 = a^2 - b^2$): 由椭圆定义 (From the definition of ellipse), 当点 P 在椭圆短轴顶点 $B(0, b)$ 时 (when point P is at the minor axis vertex $B(0, b)$ of the ellipse), $|PF_1| + |PF_2| = 2a$ 。因椭圆对称 (Due to the symmetry of the ellipse), $|PF_1| = |PF_2| = a$, 且 $|OF_1| = c$ (O 为原点 (O is the origin)). 在 $Rt \triangle BOF_1$ 中 (In $Rt \triangle BOF_1$), 由勾股定理 (by the Pythagorean theorem): $|OB|^2 + |OF_1|^2 = |BF_1|^2$, 即 (that is) $b^2 + c^2 = a^2$, 故 (so) $c^2 = a^2 - b^2$ 。



- 标准方程推导 (以 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 为例) (Derivation of Standard Equation (taking $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ as an example)): 设 $F_1(-c, 0)$, $F_2(c, 0)$, 椭圆上任意一点 $P(x, y)$, 则 (and any point $P(x, y)$ on the ellipse, then) $|PF_1| + |PF_2| = 2a$ 。代入距离公式 (Substitute the distance formula): $\sqrt{(x+c)^2 + y^2} + \sqrt{(x-c)^2 + y^2} = 2a$ 。移项得 (Transpose terms): $\sqrt{(x+c)^2 + y^2} = 2a - \sqrt{(x-c)^2 + y^2}$, 两边平方 (Square both sides): $(x+c)^2 + y^2 = 4a^2 - 4a\sqrt{(x-c)^2 + y^2} + (x-c)^2 + y^2$, 化简得 (Simplify): $4cx = 4a^2 - 4a\sqrt{(x-c)^2 + y^2}$, 即 (that is) $a\sqrt{(x-c)^2 + y^2} = a^2 - cx$, 再次平方 (Square again): $a^2[(x-c)^2 + y^2] = a^4 - 2a^2cx + c^2x^2$, 展开整理 (Expand and arrange): $(a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$, 代入 $c^2 = a^2 - b^2$ (令 $b^2 = a^2 - c^2$) (substitute $c^2 = a^2 - b^2$ (let $b^2 = a^2 - c^2$)), 得 (we get): $b^2x^2 + a^2y^2 = a^2b^2$, 两边除以 a^2b^2 (Divide both sides by a^2b^2), 即 (that is) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 。

二、例题 (Examples)

例题 1 (Example 1)

椭圆的标准方程是 $\frac{x^2}{25} + \frac{y^2}{9} = 1$, 求椭圆的焦点、顶点和离心率。 (The standard equation of an ellipse is $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find the foci, vertices and eccentricity of the ellipse.)

解析(Analysis): 第一步(Step 1),判断椭圆类型(Determine the type of ellipse): 方程为 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 形式(the equation is in the form of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$),且 $25 > 9$,故 $a^2 = 25, b^2 = 9$, 焦点在 x 轴上(the foci are on the x -axis)。

第二步(Step 2),计算 a, b, c (Calculate a, b, c): $a = 5, b = 3$, 由 $c^2 = a^2 - b^2 = 25 - 9 = 16$, 得 $c = 4 (c > 0)$ 。

第三步(Step 3),求焦点(Find the foci): $F_1(-4, 0), F_2(4, 0)$;

顶点(Vertices): x 轴上(on the x -axis) $(\pm 5, 0)$, y 轴上(on the y -axis) $(0, \pm 3)$;

离心率(Eccentricity): $e = \frac{c}{a} = \frac{4}{5}$ 。

例题 2(Example 2)

椭圆的标准方程是 $\frac{x^2}{169} + \frac{y^2}{16} = 1$, 求长轴长和短轴长。(The standard equation of an ellipse is $\frac{x^2}{169} + \frac{y^2}{16} = 1$, find the length of the major axis and the minor axis.)

解析(Analysis): 先确定 a 和 b (First determine a and b): 方程中 $a^2 = 169$ (分母较大(the larger denominator)), $b^2 = 16$, 故 $a = 13, b = 4$ 。

长轴长为(The length of the major axis is) $2a = 2 \times 13 = 26$, 短轴长为(the length of the minor axis is) $2b = 2 \times 4 = 8$ 。

补充说明(Supplementary Note): 长轴是椭圆中最长的直径,对应分母较大的变量轴,短轴为较短直径,避免混淆轴的对应关系(The major axis is the longest diameter in the ellipse, corresponding to the variable axis with the larger denominator, and the minor axis is the shorter diameter. Avoid confusing the corresponding relationship of the axes)。

三、易错点解析(Common Mistakes Analysis)

1. 焦点位置判断错误(Error in judging the position of foci): 椭圆焦点在“分母较大的变量对应的轴”上(The foci of the ellipse are on the "axis corresponding to the variable with the larger denominator"), 如 $\frac{x^2}{4} + \frac{y^2}{9} = 1$, 分母 $9 > 4$, 焦点在 y 轴上, 而非 x 轴(for example, $\frac{x^2}{4} + \frac{y^2}{9} = 1$, the denominator $9 > 4$, and the foci are on the y -axis instead of the x -axis)。
2. a, b, c 关系混淆(Confusion of the relationship between a, b , and c): 椭圆中是 $c^2 = a^2 - b^2$ (a 最大(a is the largest)), 容易与双曲线的 $c^2 = a^2 + b^2$ 混淆, 需牢记“椭圆和减, 双曲线和加”(In the ellipse, it is $c^2 = a^2 - b^2$, which is easy to be confused with the hyperbola's $c^2 = a^2 + b^2$. Remember "ellipse: difference of squares, hyperbola: sum of squares")。
3. 离心率范围记错(Misremembering the range of eccentricity): 椭圆离心率 $0 < e <$

1,若计算出 $e \geq 1$,则说明解题错误(如 $c \geq a$,违背椭圆定义 $2a > 2c$)(The eccentricity of the ellipse is $0 < e < 1$. If $e \geq 1$ is calculated, it indicates a mistake in solving the problem (e.g., $c \geq a$, which violates the ellipse definition $2a > 2c$)).

8.3 双曲线(Hyperbola)

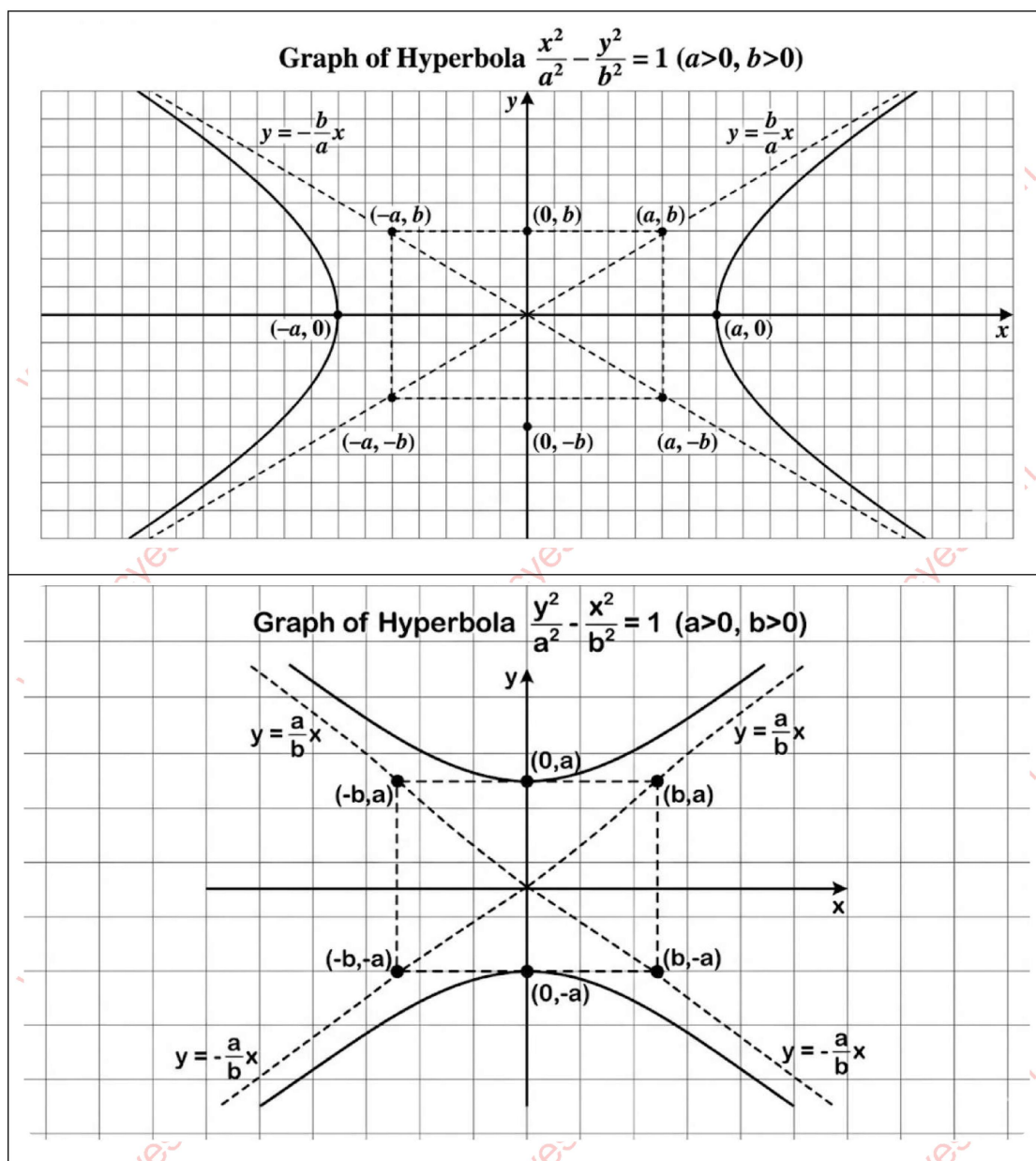
一、基础知识(Basic Knowledge)

1. 定义(Definition)

平面内与两个定点 F_1 、 F_2 (焦点)的距离差的绝对值等于常数 $2a$ ($0 < 2a < |F_1F_2| = 2c$)的点的轨迹,称为双曲线。(The set of all points in a plane where the absolute value of the difference between the distances from two fixed points F_1 and F_2 (foci) is equal to a constant $2a$ ($0 < 2a < |F_1F_2| = 2c$) is called a hyperbola.)

2. 核心公式与几何性质(Core Formulas and Geometric Properties)

标准方程 (Standard Equation)	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 (a > 0, b > 0)$	$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1 (a > 0, b > 0)$
a, b, c 关系 (Relationship between a, b, c)	$c^2 = a^2 + b^2$ (a : 实半轴(semi-transverse axis), b : 虚半轴(semi-conjugate axis), c : 半焦距(semi-focal length))	
范围(Range)	$ x \geq a, y \in \mathbf{R}$	$ y \geq a, x \in \mathbf{R}$
中心(Center)	$(0, 0)$ (原点(origin))	
对称轴 (Axis of Symmetry)	x 轴和 y 轴是对称轴,实轴长 $2a$,虚轴长 $2b$	
顶点(Vertices)	$(\pm a, 0)$	$(0, \pm a)$
焦点(Foci)	$F_1(-c, 0), F_2(c, 0)$	$F_1(0, -c), F_2(0, c)$
焦距 (Focal Length)	$ F_1F_2 = 2c (c > 0)$	
离心率(Eccentricity)	$e = \frac{c}{a} (e > 1)$, e 越大双曲线开口越开阔(the larger e is, the wider the hyperbola opens)	
渐近线(Asymptotes)	$y = \pm \frac{b}{a}x$	$y = \pm \frac{a}{b}x$



3. 公式推导(Formula Derivation)

- $c^2 = a^2 + b^2$ 推导(Derivation of $c^2 = a^2 + b^2$): 双曲线的渐近线方程由标准方程变形得到(The asymptote equation of the hyperbola is obtained by transforming the standard equation): $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$, 即 $y = \pm \frac{b}{a}x$ 。当 x 趋近于无穷大时(When x approaches infinity), 双曲线与渐近线无限接近(the hyperbola is infinitely close to the asymptote)。取双曲线上一点 $P(a, b)$ (Take a point $P(a, b)$ on the hyperbola), 代入标准方程得(substitute into the standard equation to get) $y = \pm b$, 即点 (a, b) 在渐近线 $y = \frac{b}{a}x$ 上(that is, the point (a, b) is on the asymptote $y = \frac{b}{a}x$)。连接焦点 $F_2(c, 0)$ 与点 (a, b) (Connect the focus $F_2(c, 0)$ with the point (a, b)), 距离 $|F_2P| = \sqrt{(c-a)^2 + (0-b)^2}$, 化简得(simplify to get) $|F_2P| = c$, 故 $\triangle F_2OP$ (O 为原点(O is the origin)) 中(in), $|OF_2| = c$, $|OP| = \sqrt{a^2 + b^2}$, $|F_2P| = c$, 由等腰三角形性质得(by the

property of isosceles triangle) $a^2 + b^2 = c^2$, 即(that is) $c^2 = a^2 + b^2$ 。

- 渐近线方程推导(Derivation of Asymptote Equation): 对标准方程 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, 当 $|x|$ 足够大时(when $|x|$ is sufficiently large), $\frac{x^2}{a^2} \approx \frac{y^2}{b^2}$, 变形得(transform to get) $y = \pm \frac{b}{a}x$, 即渐近线方程(描述双曲线的“趋近方向”)(which is the asymptote equation (describing the "approaching direction" of the hyperbola))。

二、例题(Example)

双曲线 $\frac{x^2}{16} - \frac{y^2}{9} = 1$ 的焦点、离心率和渐近线方程分别是什么? (What are the foci, eccentricity, and asymptote equations of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$?)

解析(Analysis): 第一步(Step 1), 判断双曲线类型(Determine the type of hyperbola): 方程为 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ 形式(the equation is in the form of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$), 焦点在 x 轴上(the foci are on the x -axis), $a^2 = 16, b^2 = 9$ 。

第二步(Step 2), 计算 a, b, c (Calculate a, b, c): $a = 4, b = 3$, 由 $c^2 = a^2 + b^2 = 16 + 9 = 25$, 得 $c = 5(c > 0)$ 。

第三步(Step 3), 求焦点(Find the foci): $F_1(-5, 0), F_2(5, 0)$;

离心率(Eccentricity): $e = \frac{c}{a} = \frac{5}{4} (e > 1, \text{符合双曲线性质(consistent with the properties of hyperbola)})$;

渐近线方程(Asymptote Equations): $y = \pm \frac{b}{a}x = \pm \frac{3}{4}x$ 。

补充验证(Supplementary Verification): 将渐近线方程代入标准方程(Substitute the asymptote equation into the standard equation), 左边为(the left side is) $\frac{x^2}{16} - \frac{(\frac{3}{4}x)^2}{9} = \frac{x^2}{16} - \frac{x^2}{16} = 0$, 与渐近线“无限接近”的定义一致(consistent with the definition of asymptote "infinitely close"), 验证正确(verification is correct)。

三、易错点解析(Common Mistakes Analysis)

1. 焦点位置判断错误(Error in judging the position of foci): 双曲线焦点在“含 x^2 或 y^2 的项的正号对应的轴”上(The foci of the hyperbola are on the "axis corresponding to the positive sign of the term containing x^2 or y^2 "), 如 $\frac{y^2}{25} - \frac{x^2}{9} = 1$, y^2 项为正(the y^2 term is positive), 焦点在 y 轴上, 而非 x 轴(the foci are on the y -axis instead of the x -axis)。

2. 渐近线方程符号或系数错误(Sign or coefficient error in the asymptote equation): $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ 的渐近线是 $y = \pm \frac{b}{a}x$, 容易写成 $y = \pm \frac{a}{b}x$ (与焦点在 y 轴的双曲线渐近线混淆)(The asymptote of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $y = \pm \frac{b}{a}x$, which is easy to be written as $y = \pm \frac{a}{b}x$ (confused with the asymptote of the hyperbola with foci on the y -axis)), 牢记“ x 轴焦点,

渐近线系数为 $\frac{b}{a}$; y轴焦点,系数为 $\frac{a}{b}$ ”(remember "foci on the x-axis: the coefficient of the asymptote is $\frac{b}{a}$; foci on the y-axis: the coefficient is $\frac{a}{b}$ ").

3. 忽略定义中“距离差的绝对值”(Ignoring the "absolute value of the difference in distances" in the definition): 若去掉“绝对值”(if the "absolute value" is removed),轨迹仅为双曲线的一支(右支或上支)(the trajectory is only one branch of the hyperbola (right branch or upper branch)),完整双曲线需满足(a complete hyperbola must satisfy)“ $||PF_1| - |PF_2|| = 2a$ ”。

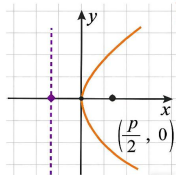
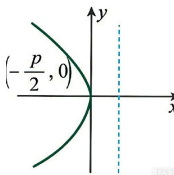
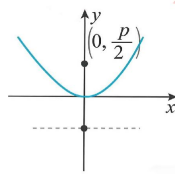
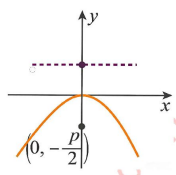
8.4 抛物线(Parabola)

一、基础知识(Basic Knowledge)

1. 定义(Definition)

平面内与一个定点 F (焦点)和一条定直线 l (准线)的距离相等的点的轨迹,称为抛物线(F 不在 l 上)。(The set of all points in a plane that are equidistant from a fixed point F (focus) and a fixed straight line l (directrix) is called a parabola (F is not on l).)

2. 核心公式与几何性质(Core Formulas and Geometric Properties)

标准方程 (Standard Equation)	$y^2 = 2px$	$y^2 = -2px$	$x^2 = 2py$	$x^2 = -2py$
				
范围(Range)	$x \geq 0, y \in R$	$x \leq 0, y \in R$	$y \geq 0, x \in R$	$y \leq 0, x \in R$
对称轴(Axis of Symmetry)	x 轴(x -axis)		y 轴(y -axis)	
顶点(Vertex)	$(0, 0)$ (原点(origin))			
焦点(Focus)	$F\left(\frac{p}{2}, 0\right)$	$F\left(-\frac{p}{2}, 0\right)$	$F\left(0, \frac{p}{2}\right)$	$F\left(0, -\frac{p}{2}\right)$
准线方程(Directrix Equation)	$x = -\frac{p}{2}$	$x = \frac{p}{2}$	$y = -\frac{p}{2}$	$y = \frac{p}{2}$
离心率 (Eccentricity)	$e = 1$ (抛物线离心率恒为 1(the eccentricity of a parabola is always 1))			

3. 公式推导(Formula Derivation)

- 以 $y^2 = 2px (p > 0)$ 为例推导(Derivation taking $y^2 = 2px (p > 0)$ as an example):
设焦点 $F\left(\frac{p}{2}, 0\right)$, 准线 $l: x = -\frac{p}{2}$, 抛物线上任意一点 $P(x, y)$ (Let the focus be $F\left(\frac{p}{2}, 0\right)$, the directrix $l: x = -\frac{p}{2}$, and any point $P(x, y)$ on the parabola)。根据定义(According to the definition), 点 P 到 F 的距离等于到 l 的距离(the distance from point P to F is equal to the distance to l), 即 $|PF| = d$ (d 为 P 到 l 的距离(d is the distance from P to l))。代入距

离公式(Substitute the distance formula): $\sqrt{\left(x - \frac{p}{2}\right)^2 + (y - 0)^2} = \left|x + \frac{p}{2}\right|$, 两边平方
(Square both sides): $\left(x - \frac{p}{2}\right)^2 + y^2 = \left(x + \frac{p}{2}\right)^2$, 展开化简(Expand and simplify): $x^2 - px + \frac{p^2}{4} + y^2 = x^2 + px + \frac{p^2}{4}$, 消去同类项得(Eliminate like terms to get): $y^2 = 2px$, 即标准方程(which is the standard equation)。其中 p 的几何意义(The geometric meaning of p here): 焦点到准线的距离(the distance from the focus to the directrix)($|OF| = \frac{p}{2}$, 准线到原点距离 $\frac{p}{2}$, 故 p = 焦点到准线距离(the distance from the directrix to the origin is $\frac{p}{2}$, so p = the distance from the focus to the directrix))。

二、例题(Example)

顶点在原点,焦点在 x 轴上,并且经过点 $P(2, 4)$ 的抛物线方程是() (The equation of the parabola with vertex at the origin, focus on the x -axis, and passing through the point $P(2, 4)$ is ())

A. $y^2 = 8x$ B. $y^2 = 4x$ C. $x^2 = 2y$ D. $x^2 = \frac{1}{2}y$

解析(Analysis): 第一步(Step 1),确定抛物线类型(Determine the type of parabola): 焦点在 x 轴上,顶点在原点(focus on the x -axis and vertex at the origin),故标准方程为(so the standard equation is) $y^2 = 2px$ 或 $y^2 = -2px$ ($p > 0$),排除选项 C、D(焦点在 y 轴上(foci on the y -axis))。

第二步(Step 2),判断开口方向(Judge the opening direction): 点 $P(2, 4)$ 在第一象限(the point $P(2, 4)$ is in the first quadrant), $x = 2 > 0$, 故抛物线开口向右(so the parabola opens to the right), 方程为(the equation is) $y^2 = 2px$ ($p > 0$)。

第三步(Step 3),代入点求 p (Substitute the point to find p): 将 $P(2, 4)$ 代入方程得(substitute $P(2, 4)$ into the equation to get) $4^2 = 2p \times 2$, 即 $16 = 4p$, 解得(solve for) $p = 4$ 。

第四步(Step 4),确定方程(Determine the equation): $y^2 = 2 \times 4x = 8x$, 对应选项 A(corresponding to option A)。

补充思路(Supplementary Idea): 若用定义验证(If verified by definition), 焦点 $F\left(\frac{p}{2}, 0\right) = (2, 0)$, 准线 $x = -2$, 点 $P(2, 4)$ 到 F 的距离 $|PF| = \sqrt{(2-2)^2 + (4-0)^2} = 4$, 到准线的距离 $d = 2 - (-2) = 4$, $|PF| = d$, 符合抛物线定义(consistent with the definition of parabola), 验证正确(verification is correct)。

三、易错点解析(Common Mistakes Analysis)

1. 焦点坐标与准线方程符号错误(Sign error of focus coordinates and directrix equation): 如 $y^2 = -2px$ ($p > 0$), 焦点是 $\left(-\frac{p}{2}, 0\right)$, 准线是 $x = \frac{p}{2}$, 容易将符号写反(与 $y^2 = 2px$ 混淆)(For example, for $y^2 = -2px$ ($p > 0$), the focus is $\left(-\frac{p}{2}, 0\right)$ and the directrix

is $x = \frac{p}{2}$, which is easy to have reversed signs (confused with $y^2 = 2px$)), 牢记“开口方向决定符号: 向右/向上为正, 向左/向下为负”(remember "the opening direction determines the sign: right/up is positive, left/down is negative")。

2. p 的几何意义理解错误(Misunderstanding of the geometric meaning of p): p 是焦点到准线的距离(p is the distance from the focus to the directrix), 而非焦点到顶点的距离(焦点到顶点距离为 $\frac{p}{2}$)(not the distance from the focus to the vertex (the distance from the focus to the vertex is $\frac{p}{2}$)), 解题时若直接将焦点坐标设为 $(p, 0)$, 会导致错误(if the focus coordinate is directly set to $(p, 0)$ when solving the problem, it will lead to an error)。

3. 标准方程形式混淆(Confusion of standard equation forms): 如焦点在 y 轴上的抛物线方程是 $x^2 = 2py$, 而非 $y^2 = 2px$ (For example, the equation of a parabola with focus on the y -axis is $x^2 = 2py$, not $y^2 = 2px$), 需根据“ x^2 对应 y 轴焦点, y^2 对应 x 轴焦点”区分(distinguish according to " x^2 corresponds to focus on the y -axis, y^2 corresponds to focus on the x -axis")。

圆锥曲线核心性质对比总结

(Comparison and Summary of Core Properties of Conic Sections)

曲线类型 (Type of Curve)	定义核心(Core of Definition)	标准方程关键特征(Key Features of Standard Equation)	$a, b, c/p$ 关系 (Relationship between $a, b, c/p$)	离心率 e (Eccentricity e)	核心区别(Core Differences)
圆(Circle)	到定点距离为定长 (The set of points equidistant from a fixed point (center) to a fixed length (radius))	平方和等于常数(Sum of squares equals a constant)(($x - a)^2 + (y - b)^2 = r^2$)	r 为半径(r is the radius),无 a, b, c 关系(no relationship between a, b, c)	$e = 0$ (特殊值 (special value))	封闭曲线(closed curve),离心率为0(eccentricity is 0),无焦点差异(no difference in foci)
椭圆(Ellipse)	距离和为常数($2a > 2c$)(The set of points where the sum of distances from two fixed points (foci) is a constant($2a > 2c$))	平方和为1(Sum of squares is 1)(($\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$)	$c^2 = a^2 - b^2$ ($a > b$)	$0 < e < 1$	封闭曲线(closed curve), e 越小越圆(the smaller e is, the more circular it is)
双曲线(Hyperbola)	距离差的绝对值为常数($2a < 2c$)(The set of points where the absolute value of the difference in distances from two fixed points (foci) is a constant($2a < 2c$))	平方差为1(Difference of squares is 1)(($\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$)	$c^2 = a^2 + b^2$ ($c > a$)	$e > 1$	开放曲线(open curve),有渐近线(has asymptotes)
抛物线(Parabola)	到定点与定直线距离相等(The set of points equidistant from a fixed point (focus) and a fixed straight line (directrix))	单变量平方等于一次项(Square of a single variable equals a linear term)(($y^2 = 2px$)	p 为焦点到准线距离(p is the distance from focus to directrix)	$e = 1$ (定值 (fixed value))	开放曲线(open curve),无 a, b, c (no a, b, c),离心率恒为1(eccentricity is always 1)